

Solutions for Chapter 15

Atmospheric and Oceanic Fluid Dynamics

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- 15.2 (a) Consider an ocean with a flat surface, forced only by radiation. Suppose that the net radiation at the surface falls from about $+100 \text{ W m}^{-2}$ at the equator to -100 W m^{-2} at the pole, and that associated heating $\dot{Q}$ varies in vertical with an e-folding scale of 10 metres with a volume integral that is zero. Is such a forcing sufficient to account for the observed dissipation of kinetic energy in the ocean, which is about 1 kW km^{-3} ?
- (b) Now let a wind blow over the ocean, providing an average stress of 0.05 N m^{-2} . Estimate the kinetic energy dissipation that this will lead to.
- (c) Estimate the rate of heating (in degrees per day) associated with a kinetic energy dissipation rate of 1 kW km^{-3} ?

There are two aspects to this problem that might cause difficulty. One is just the manipulation of all the units, and the second is the more conceptual one of setting up the problem properly.

- (a) The derivation in AOFD begins with a thermodynamic equation of the form $Db/Dt = Q_b$. Here it is convenient to use a thermodynamic equation of the form

$$\rho_0 c_p \frac{DT}{Dt} = Q_T, \quad (\text{S15.1})$$

because the source term, Q_T , is in Watts/m³. Given that $b = -g\delta\rho/\rho_0$ and the equation of state $\rho = \rho_0(1 - \beta_T(T - T_0))$ we obtain

$$\frac{Db}{Dt} = \left(\frac{g\beta_T}{\rho_0 c_p} \right) Q_T \quad (\text{S15.2})$$

Similarly, the units of kinetic energy dissipation in the book, ε , are in units of W kg^{-1} do not include a factor of ρ_0 , and need to do so to be in units of W m^{-2} . Let us define $\varepsilon' = \rho\varepsilon$, where ε' has units of W m^{-2} . Equation (15.19) of AOFD, namely

$$\langle zQ_b \rangle = -\langle \varepsilon \rangle, \quad (\text{S15.3})$$

then becomes

$$\langle yzQ_T \rangle = -\langle \varepsilon' \rangle \quad (\text{S15.4})$$

where $\gamma = g\beta_t/c_p \approx 5 \times 10^{-7} \text{ m}^{-1}$, and if the angle brackets are taken here to denote an average then both the left- and right-hand sides of (S15.4) are in units of W m^{-3} . (Note that γz is a dimensionless quantity.) We are given that $\varepsilon' = 10^{-6} \text{ W m}^{-2}$.

To simplify matters, let us simply assume that the heating and cooling are distributed so that there is a heating rate of 100 W m^{-3} between 9 m and 10 m depth and a cooling of 100 W m^{-3} within 1 m of the surface, and elsewhere the heating is zero. (This differs slightly from the question as stated, but the differences will be small and quantitative, and not large or qualitative.) The left-hand side of (S15.4) is then, approximately

$$\langle \gamma z Q_T \rangle \sim 5 \times 10^{-7} \times -10 \times 100 \times (2/5000) \sim 2^{-7}. \quad (\text{S15.5})$$

The factor of $((2/5000))$ accounts for the fact that we need to compute the average energy generation and dissipation over an ocean that is about 5000 m deep, but the heating is only non-zero over two 1 m intervals. This calculation implies that heating in the upper ocean can not quite supply enough energy to account for the average dissipation of kinetic energy in the ocean. *However, the calculation is not exact* and, furthermore, much of the kinetic energy dissipation actually occurs in the mixed layer itself, and here it is much larger than 10^{-6} W m^{-3} . The take home message is that energetic calculations are difficult, and the results tend to be a little ambiguous.

If the heating were distributed over a thickness greater than 1 m, but still at an average depth of 10 m, the calculation would be essentially unchanged. For example, suppose the heating occurred uniformly between 5 m and 10 m depth. The heating rate would then be 10 W m^3 (instead of 100 W m^3) and the averaging factor would be $(10/5000)$.

However, there is one message that rings out clearly. Suppose that the heating mainly occurred at 1000 m depth, and the cooling remained at the surface. Then the above generation rate would be multiplied by 100, which would yield a number far greater than the average kinetic energy dissipation rate in the ocean.

(b) With an imposed stress, the horizontal equation becomes

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla\phi + \frac{1}{\rho_0} \frac{\partial \boldsymbol{\tau}}{\partial z} + \nu \nabla^2 \mathbf{u}. \quad (\text{S15.6})$$

If the imposed stress is zero except at the surface, then on taking the dot product of (S15.6) and integrating in the vertical, we see that the stress provides an energy input, E_τ , of

$$E_\tau = \boldsymbol{\tau} \cdot \mathbf{u}(z = 0) \quad (\text{S15.7})$$

with units of $\text{J m}^{-2} \text{ s}^{-1}$. To estimate a value, let us suppose that the surface current is 0.1 m s^{-1} and in the direction of the wind. The average energy input per unit area is then $E_\tau \approx 5 \times 10^{-3} \text{ W m}^{-2}$. If we spread this out over the depth of the ocean (5000 m) it translates to a rate of 10^{-6} W m^{-3} , similar to the observed kinetic

energy dissipation rate. Again, however, the calculation must be regarded as very inexact, and more careful calculations give a somewhat higher figure for the wind work.

The total area of the ocean is about $3 \times 10^{14} \text{ m}^2$, so the above figures represent a total amount of wind work of $1.5 \times 10^{12} \text{ W}$. Typically, the useful unit of power for the ocean as a whole is the terrawatt, where $1 \text{ TW} = 10^{12} \text{ W}$.

(c) To obtain the Joule heating rate of the ocean we can use the equation

$$\rho_0 c_p \frac{DT}{Dt} = Q_J, \quad (\text{S15.8})$$

where Q_J is the Joule heating, that is the heating of the fluid due to the kinetic energy dissipation. Note the Q_J is much *smaller* than the actual heating rate of the ocean, Q_T , sometimes called the ‘caloric’ heating: most of this heating is not available for conversion to kinetic energy. Putting in numbers, we find

$$\frac{dT}{dt} \approx \frac{Q_J}{\rho_0 c_p} \approx \frac{10^{-6}}{10^3 \cdot 4000} \approx 2 \times 10^{-13} \text{ K s}^{-1} = 2 \times 10^{-8} \text{ K day}^{-1}. \quad (\text{S15.9})$$

(This seems too small to me. Can anyone see a mistake?)

15.3 Consider a model of sideways convection in which the boundary condition at the top is the relaxation, or Haney, condition $b_z = A(b^*(y) - b)$ where A (a constant) and $b^*(y)$ are given, b is the buoyancy, proportional to the temperature, the other boundaries are insulating, and the flow is statistically steady.

- (a) Is it still the case that on average the fluid is heated (i.e., there is a heat flux into the fluid through the upper surface) where it is already warm? If so, how may this be reconciled with the intuition that if the ocean surface is anomalously warm it will cool by way of a heat flux from the ocean to the atmosphere?
- (b) Show that if $A < 0$ a statistically steady state cannot be reached.
- (c) Suppose that b^* varies monotonically with latitude. Show that if κ is non-zero, the average surface temperature gradient will be less than that of b^* .

(a) We take the equation of motion to be

$$\frac{Db}{Dt} = \kappa \nabla^2 b. \quad (\text{S15.10})$$

where $\kappa > 0$. The given boundary condition at the surface tells us that the surface flux is given by

$$\text{Flux} = \kappa \frac{\partial b}{\partial z} = \kappa A(b^*(y) - b), \quad (\text{S15.11})$$

where without loss of generality we may assume that the average of b^* is zero.

Multiply (S15.10) by b and integrate over the domain to give the buoyancy variance equation,

$$\frac{1}{2} \frac{d \langle b^2 \rangle}{dt} = \kappa \left[\left. b \frac{\partial b}{\partial z} \right|_{z=0} - \langle |\nabla b|^2 \rangle \right], \quad (\text{S15.12})$$

where we assume that the normal derivative of b vanishes on all the surfaces except the top one ($z = 0$), and an overbar denotes a horizontal integral. Thus, in a statistically steady state

$$\overline{b \frac{\partial b}{\partial z}} \Big|_{z=0} = \langle |\nabla b|^2 \rangle. \quad (\text{S15.13})$$

The right-hand side is positive definite, so that b and $\partial b / \partial z$ must be positively correlated. Because the flux is $\kappa \partial b / \partial z$, regardless of the boundary condition given by (S15.11), then the positive correlation implies that there is on average a flux of buoyancy into the fluid, $\partial b / \partial z > 0$, when the fluid is buoyant, $b > 0$.

There is no difficulty in the reconciliation, because the statement refers to average conditions. In the real world it implies, for example, that there is a flux into the ocean at low latitudes (where the ocean is warm) and a flux out of it at high latitudes where the ocean is cold. If there is a warm sea surface temperature anomaly in some particular region for some period of time, then there may be a flux out of the ocean, but this does not contradict the result.

(b) Let us write the equation of motion as

$$\frac{\partial b}{\partial t} + \nabla \cdot (\mathbf{v}b) = \nabla \cdot \mathbf{F} \quad (\text{S15.14})$$

where $\mathbf{F}$ is the diffusive flux, given by $\mathbf{F} = \kappa \nabla b$. Integrate over the domain, applying no normal-flow boundary conditions so that the advective term vanishes, and obtain

$$\frac{\partial \langle b \rangle}{\partial t} = \overline{F^z(0)} = \overline{\kappa A(b^*(y) - b)}. \quad (\text{S15.15})$$

The right-hand side represents the vertical flux at the surface $z = 0$ and we assume that there is no diffusive flux at the other boundaries. If $A < 0$ then the equation will be unstable, for there is a potential for exponential growth. Informally, the buoyancy equation at the surface is of the form $\partial b / \partial t \propto \kappa A(b - b^*)$, which is unstable.

(c) Using (S15.11) in (S15.13) gives

$$A \overline{b(b^*(y) - b)} = \langle |\nabla b|^2 \rangle, \quad (\text{S15.16})$$

where A is now assumed positive. The right-hand side is positive definite, so that, assuming for simplicity a two-dimensional ocean (in y and z),

$$\int b b^* dy > \int b^2 dy \quad (\text{S15.17})$$

If b^* varies monotonically with y , then this inequality is only satisfied if the gradient of b across the domain is *less* than the gradient of b^* . Thus, the diffusivity is acting to reduce the surface buoyancy gradient, and therefore transfer heat from regions of high temperature to regions of low temperature, consistent with the second law of thermodynamics.